

Pelvic architecture shapes load-dependent sacroiliac motion

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Abstract

Pelvic stability depends on transmitting load through joints that permit only small movements. How does anatomical variation shape these movements when the load path changes? We combined imaging-derived anatomy from 278 pelves (150 female, 128 male) with finite-element analysis of five standardized loads and matched geometry–bone-density variants. Redistributing the same 800 N acetabular force from bilateral to unilateral support increased median sacroiliac translation from 1.058 to 1.627 mm and bilateral translation asymmetry from 0.065 to 1.602 mm. Three localized force pairs produced distinct motion-component patterns, rather than a common increase in transverse motion. Positive age-adjusted female translation differences attenuated after including pelvic size and dimensions; fully adjusted coefficients ranged from -0.031 to +0.013 mm, with all confidence intervals spanning zero. All 20 sex-specific volume slopes were negative after multiplicity correction. Retaining individual geometry with a population-reference bone-density field reproduced full-model translation with RMSE no greater than 0.028 mm across loads. These controlled comparisons show how pelvic architecture and load distribution jointly shape sacroiliac motion, and why sex alone provides an incomplete mechanical description. The findings identify anatomical and loading contrasts for experimental testing; their quantitative scope is passive static mechanics with fixed soft-tissue properties.

Keywords: sacroiliac joint; pelvic functional anatomy; pelvic biomechanics; sexual dimorphism; pelvic architecture; load transfer; finite-element analysis

1 Introduction

The pelvis must transmit substantial loads through joints that move only slightly. This combination makes the sacroiliac joint (SIJ) an informative site at which to study how skeletal

architecture organizes movement: small changes in relative motion can reveal how a load travels through the pelvic ring. Articular geometry, ligamentous restraint and muscular compression together contribute to pelvic stability (Vleeming et al., 2012; Snijders et al., 1993; Hammer et al., 2013). Yet a description of how much the SIJ moves is incomplete without specifying where the load acts and which anatomy carries it.

This problem also changes how pelvic sexual dimorphism should be interpreted mechanically. Female and male pelves differ in dimensions relevant to both load transfer and obstetric anatomy. A difference in joint motion between these groups could reflect overall size, particular dimensions, or anatomical features beyond the measurements available. Distinguishing these possibilities is more informative than assigning each sex a single degree of pelvic mobility. The relationship between locomotor support and obstetric accommodation gives this question biological importance (Pavličev et al., 2020); mechanical models can address its structural basis, while simulations of childbirth additionally require fetal contact, soft tissues and changing boundary conditions (Chen and Grimm, 2021).

Previous population modeling has linked SIJ morphology to its mechanical environment (Henyš and Hammer, 2025). The next question is how the same pelvis responds when the load path changes, and whether the resulting differences among individuals follow pelvic architecture or bone-density variation. Addressing this experimentally across many anatomies is difficult: loading and tissue properties cannot readily be varied independently in vivo. Matched computational comparisons provide a way to hold these factors constant in turn. Their value lies in the controlled comparisons they enable, alongside the anatomical diversity they retain.

Here, we combine imaging-derived anatomy from 278 pelves with five standardized load configurations and three matched geometry–density variants, using the BoneDat workflow (Henyš and Kuchař, 2025). Bilateral and unilateral acetabular loading redistribute the same total force. Three further force pairs probe the ring at sites motivated by obstetric anatomy; they are independent mechanical probes, not successive stages of labor. This design links three scales of explanation: the load applied to the ring, the motion expressed at its joints, and the anatomical differences among individuals.

We organize the analysis around four hypotheses. **H1:** shifting from bilateral to unilateral support increases SIJ motion and bilateral asymmetry. **H2:** the two transverse force pairs increase the absolute mediolateral component and reduce the craniocaudal component relative to bilateral support; the anteroposterior pair is assessed separately. **H3:** greater female translation

under the localized force pairs is attenuated when pelvic size and dimensions are included. **H4:** retaining individual geometry while standardizing the bone-density field largely preserves both between-subject variation and individual joint responses. Volume–motion associations provide a complementary, exploratory test of the role of pelvic size. Together, these comparisons ask whether pelvic mobility is better understood as a response specific to anatomy and loading than as a fixed characteristic of a sex or an individual.

2 Materials and methods

2.1 Cohort and provenance

The precomputed cohort comprised 278 pelvic models (150 female, 128 male), aged 16–91 years (mean $\pm$ SD: 53.9 ± 19.3 years), using the BoneDat morphology workflow (Henyš and Kuchař, 2025). Demographic, morphometric and size tables were joined by subject identifier. Eight morphometric variables and the kinematic outcomes were complete for the included cohort (Table 1). The previously recorded ethical approval is University Hospital Hradec Králové, reference 202411IO3P, with waiver of informed consent. Documentation of the cohort selection and its overlap with the earlier 281-model study (Henyš and Hammer, 2025) is required to distinguish reuse of imaging from independent validation.

Table 1: Demographic and morphometric characteristics. Values are mean $\pm$ SD; dimensions are measured from mapped landmarks.

Variable	Overall	Female	Male	Missing
N	278	150	128	0
Age (years)	53.90 ± 19.33	53.36 ± 19.74	54.52 ± 18.91	0
Inlet AP diameter (mm)	117.04 ± 10.52	121.04 ± 10.02	112.34 ± 9.07	0
Biacetabular width (mm)	125.81 ± 8.53	129.08 ± 8.21	121.98 ± 7.22	0
Biischiadic width (mm)	104.27 ± 9.33	109.87 ± 7.13	97.71 ± 7.06	0
Bituberous width (mm)	132.24 ± 9.15	136.46 ± 8.29	127.30 ± 7.49	0
Iliopectineal eminence width (mm)	102.38 ± 7.75	106.20 ± 7.02	97.90 ± 5.96	0
Pelvic inlet transverse (mm)	131.58 ± 8.88	134.85 ± 8.59	127.75 ± 7.61	0
Sacral width (mm)	117.08 ± 6.08	116.75 ± 5.67	117.47 ± 6.54	0
Subpubic angle (deg)	78.12 ± 10.12	85.80 ± 6.14	69.11 ± 5.26	0
Global scale factor (-)	0.96 ± 0.06	0.92 ± 0.05	1.00 ± 0.05	0
Total pelvic volume (cm ³)	823.58 ± 149.32	736.79 ± 105.99	925.28 ± 126.99	0
Total pelvic surface (cm ²)	1568.63 ± 169.26	1479.09 ± 133.64	1673.57 ± 144.80	0

2.2 Finite-element model

The archived simulation metadata specify DOLFINx 0.10, 166,808 cells and 48,291 mesh nodes. Small-strain linear elasticity was solved with PETSc on a shared reference domain. Subject anatomy was represented by $\mathbf{x} = \mathbf{X} + \phi_i(\mathbf{X})$; the displacement map was interpolated using radial basis functions (RBFs). Spatial gradients and integration measures were pulled back to the reference domain (Bonet and Wood, 1997). This representation approximates individual anatomy on a shared discretization; it does not guarantee exact cortical boundaries or invertibility without Jacobian checks.

The implemented bone-modulus law was

$$E/\text{MPa} = \begin{cases} 2398, & \rho_{\text{ash}} \leq 0.486, \\ 10200\rho_{\text{ash}}^2, & \rho_{\text{ash}} > 0.486, \end{cases} \quad \rho_{\text{ash}} = (\rho + 0.09)/1.14, \quad (1)$$

where numerical density values are in g cm^{-3} and ρ denotes the mapped hydroxyapatite-density input. This implemented approximation has a constant low-density branch, which suppresses modulus variation below its threshold; it must not be interpreted as an unrestricted test of bone quality (Keller, 1994). Bone Poisson ratio was 0.30. Both SIJ cartilage and pubic symphysis used $E = 1.0$ MPa and $\nu = 0.45$, as recorded in all three simulation archives. These nominal choices were not calibrated to individual subjects.

Ligaments were assembled as linear spring stiffness matrices with a geometric stiffness contribution from prescribed pretension and a pretension force vector. The implementation does not update a tension-only active set during the static solve. Bilateral anterior SIJ, posterior SIJ, interosseous, sacrospinous and sacrotuberous bundles and the pubic bundle were included. Per-bundle stiffnesses were respectively 1400, 5600, 200, 3000, 3000 and 1000 N/mm; total pretension was 20 N per bundle. Each bundle’s parameters were divided among its segments. Bone–cartilage interfaces were conforming; frictional contact and muscle activation were not modeled.

2.3 Loads and constraints

The superior S1 facet was constrained using a surface penalty $\gamma = 10^6$ N/mm³ (Appendix A). Table 2 gives force vectors in the simulation’s global coordinates; these must be distinguished from the subsequently constructed pelvis-aligned reporting frame. SP2leg applies +400 N ver-

tically to each acetabulum, and SP1leg applies +800 N to the right acetabulum. LAB1 applies opposing ML forces of ± 400 N to the internal ring surfaces; LAB2 applies the same signed pair to the ischial tuberosities; LAB3 applies opposing AP forces of ± 400 N at the pubis and sacro-coccygeal junction. Inspection of the archived template places both left application patches at greater global x than their right counterparts; thus both ML pairs point outward in that template, consistent with outward loading at both sites. We name the cases by application site because a force direction does not establish the direction of joint motion.

Surface tractions were normalized by mapped surface area, giving identical prescribed force resultants across subjects. LAB force pairs have zero net force, whereas standing cases have an 800 N net vertical force balanced by the sacral constraint. Equal sums of force magnitudes do not imply equal moments, work, or physiological demand. The analyses compare responses to these specified load systems, not force-normalized compliance across equivalent activities. All load cases include the prescribed ligament pretension.

Table 2: Standardized loads in global simulation coordinates. LAB1–LAB3 are independent mechanical proxies.

Load case	Mechanical proxy	Applied total forces	Resultant
SP2leg	Bilateral standing	[0,0,+400] N on right AC notch and [0,0,+400] N on left AC notch	800 N cranio-caudal
SP1leg	Unilateral standing	[0,0,+800] N on right AC notch	800 N cranio-caudal
LAB1	Internal ring force pair	[+400,0,0] N on ring_contact_left and [-400,0,0] N on ring_contact_right	0 N net; bilateral ML pair
LAB2	Ischial force pair	[+400,0,0] N on left_ischium_tuber and [-400,0,0] N on right_ischium_tuber	0 N net; bilateral ML pair
LAB3	Outlet AP force pair	[0,-400,0] N on pubis_ins and [0,+400,0] N on SCJ	0 N net; AP pair

2.4 Reference configuration and SIJ kinematics

The kinematic analysis compares each unloaded mapped surface $\mathbf{x}_i^0 = \mathbf{X} + \phi_i$ with its loaded surface $\mathbf{x}_{i\ell}^1 = \mathbf{x}_i^0 + \mathbf{u}_{i\ell}$. Stored FE displacement arrays were reprocessed without rerunning equilibrium solves. Unloaded-to-unloaded checks were below 10^{-8} in degrees and mm for every subject. Node ordering was checked against all five archived reference displacement fields. Geometry and displacement were evaluated with the same RBF parameters used by the original workflow. Checkpoints preserve subject row indices consistently across the three variants. Pairing follows the common indexed shape/density inputs and the archived processing order; the auxiliary archives do not contain an independent subject-ID manifest.

Relative ilium-to-sacrum motion was estimated using trimmed Kabsch registration of opposing surface regions within 5 mm, with a minimum of 80 nodes per region. Before registration, the inverse least-squares affine sacral map was applied to all loaded points. This removes sacral

affine strain as well as rigid drift; the resulting quantities are affine-corrected relative motion, rather than unmodified physical joint displacement. Sensitivity to rigid-only drift removal remains necessary for external validation.

Motion was expressed in a pelvis-aligned ML/AP/CC frame, not a local articular normal/tangent frame. Fixed-axis Cardan xyz angles satisfy $R = R_z R_y R_x$. The bilateral mean of the Euclidean norms of these angle triplets is denoted $|\boldsymbol{\theta}|$; this is a small-angle summary, not the exact invariant rotation angle. Translation $|\mathbf{d}|$ is the bilateral mean of the norms of relative displacements evaluated at the sacral ROI centroid. Component summaries are bilateral means of absolute values. In particular, $|\theta_x|$ does not distinguish nutation from counternutation, and $|d_{ML}|$ is not a measurement of joint gapping. Translation-magnitude asymmetry is $||\mathbf{d}_L|| - ||\mathbf{d}_R||$.

2.5 Matched variants and variance ratios

The full variant uses subject-specific geometry and bone density. The shape-only variant uses subject geometry and the spatially varying pointwise population-median density field; it is not homogeneous bone. The material-only variant uses subject density on the template geometry. Cartilage and ligament parameters remain nominal in all variants.

For each load and endpoint we calculated $100V_{shape}/V_{full}$ and $100V_{material}/V_{full}$, with 2000 paired subject bootstrap samples (Efron and Tibshirani, 1994). These non-additive ratios can exceed 100% and neither estimate causal variance fractions nor separate geometry-material interactions. Shape/full agreement was also assessed by RMSE, MAE and identity-line $R^2 = 1 - \sum_i (Y_{shape,i} - Y_{full,i})^2 / \sum_i (Y_{full,i} - \bar{Y}_{full})^2$.

2.6 Statistical analysis

The hypotheses organize a retrospective analysis and were not preregistered. Distributions are summarized by medians and interquartile ranges. Paired component contrasts use $\Delta_{i\ell,k} = |d_k|_{i\ell} - |d_k|_{i,SP2leg}$, with 3000 bootstrap samples for median confidence intervals and two-sided exact binomial sign tests of the median contrast (zero differences excluded). Wilcoxon signed-rank statistics are retained in the machine-readable output as a sensitivity diagnostic, but are not used for the primary inference because asymmetric differences can yield a signed-rank result that disagrees with the direction of the median. Benjamini-Hochberg correction was applied to the 12 component comparisons (Benjamini and Hochberg, 1995). H1 additionally uses paired tests

for rotation magnitude, translation magnitude and translation asymmetry (a separate three-test family).

For each LAB case, nested OLS models used HC3 covariance (MacKinnon and White, 1985):

$$M0 : Y = \beta_0 + \beta_s \text{Female} + \beta_a \text{Age}_z + \epsilon, \quad (2)$$

$$M1 : Y = M0 + \beta_v \log(V)_z, \quad (3)$$

$$M2 : Y = M1 + \beta_{AP} \text{AP}_z + \beta_b \text{Biischadic}_z + \beta_p \text{Subpubic}_z. \quad (4)$$

Here model addition denotes addition to the linear predictor. Female is coded 1 and male 0; continuous predictors are standardized using cohort sample SDs. M2 contains no sex–volume interaction. FDR correction covers its six sex coefficients (two outcomes by three loads). M0/M1 and coefficient attenuation are descriptive. VIFs quantify collinearity without treating a threshold as proof of identifiability. The triad describes inlet depth, interspinous midpelvic width and the anterior pubic arch; it was selected for this retrospective analysis. Pooled and sex-stratified Spearman correlations are exploratory, not evidence of mediation.

Pooled OLS models include load, sex, age, log volume, the morphometric triad, load–volume interactions and a sex–volume interaction, with covariance clustered by the 278 subjects. A main volume coefficient therefore refers to male SP2leg, not an overall pooled slope. FDR is applied within each outcome’s pooled coefficients. Robust Wald statistics are not converted to partial variance-explained measures.

Exploratory load-specific models regress $\log Y$ on $\log V$, sex, their interaction, age and the triad. Male and female slopes are β_v and $\beta_v + \beta_{v:s}$; HC3 covariance supplies 95% Wald intervals. FDR covers 20 sex-specific slope tests, with a separate family of ten interaction tests. Failure to reject an interaction is not evidence of equal slopes. The alternative $\log(\text{scale})$ model is an algebraic reparameterization because $\log V = 3 \log(\text{scale}) + \log V_{\text{template}}$; it is not an independent robustness test. Sample confidence intervals do not include uncertainty in loads, constitutive choices, segmentation or boundary conditions.

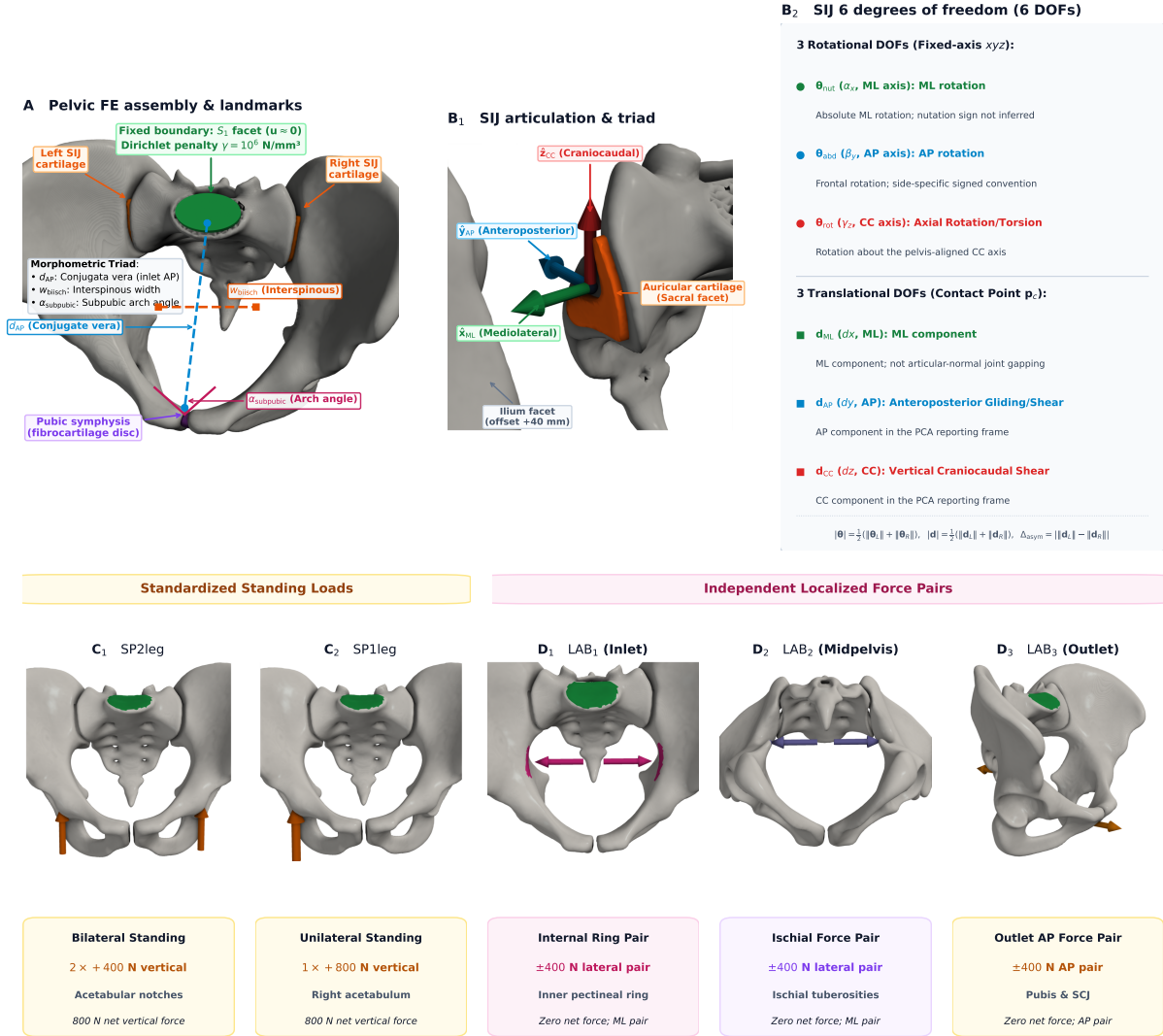


Figure 1: Model anatomy and load definitions. (A) Pelvic assembly with schematic morphometric overlays; their drawn locations illustrate dimensions and are not quantitative landmark measurements. Green denotes the S1 constraint, orange the SIJ cartilage and purple the symphysis. (B) Exploded articulation and pelvis-aligned coordinate definitions; arrows illustrate axes, not measured trajectories. Rotations describe ilium relative to sacrum. (C) Bilateral and unilateral acetabular loads. (D) Independent ring, ischial and AP force pairs. Arrows represent applied global force directions; they do not demonstrate joint opening. The penalty approximates a fixed boundary.

3 Results

3.1 Anatomical diversity across matched mechanical comparisons

The analysis includes 278 subjects and 1390 subject-load observations in the full variant, with corresponding records in both auxiliary variants. Cohort demographics and dimensions are unchanged (Table 1). Each anatomy contributes matched responses to all five loads and all three variants.

3.2 Unilateral support amplifies asymmetry at unchanged total force

Under SP2leg and SP1leg, median rotation norms were 0.956° and 2.040° , and median translations were 1.058 and 1.627 mm, respectively. Median bilateral translation asymmetry was 0.065 mm in SP2leg and 1.602 mm in SP1leg (Fig. 2). The paired SP1leg-minus-SP2leg median difference in rotation norm was $+1.048$ (95% CI $+1.004$ to $+1.102$) degrees, $q < 0.001$. The paired SP1leg-minus-SP2leg median difference in translation magnitude was $+0.582$ (95% CI $+0.559$ to $+0.603$) mm, $q < 0.001$. The paired SP1leg-minus-SP2leg median difference in translation asymmetry was $+1.509$ (95% CI $+1.445$ to $+1.568$) mm, $q < 0.001$. Redistributing the same 800 N total force therefore increased motion and, most prominently, bilateral asymmetry.

Table 3: SIJ motion summaries across 278 subjects. Values are median [interquartile range]; component columns are absolute magnitudes.

Load case	N	Rotation magnitude (deg)	Translation magnitude (mm)	AP translation component (mm)	ML translation component (mm)	CC translation component (mm)
SP2leg	278	0.956 [0.813, 1.109]	1.058 [0.871, 1.207]	0.366 [0.241, 0.482]	0.083 [0.054, 0.123]	0.952 [0.818, 1.112]
SP1leg	278	2.040 [1.765, 2.289]	1.627 [1.430, 1.869]	0.440 [0.357, 0.520]	0.235 [0.164, 0.345]	1.428 [1.213, 1.644]
LAB1	278	0.221 [0.188, 0.268]	0.154 [0.121, 0.190]	0.060 [0.044, 0.079]	0.077 [0.063, 0.092]	0.105 [0.073, 0.144]
LAB2	278	0.724 [0.649, 0.812]	0.255 [0.204, 0.328]	0.176 [0.099, 0.262]	0.134 [0.115, 0.154]	0.075 [0.049, 0.118]
LAB3	278	0.844 [0.672, 1.057]	0.574 [0.448, 0.707]	0.372 [0.260, 0.515]	0.092 [0.069, 0.115]	0.387 [0.326, 0.477]

3.3 Load application site selects distinct motion components

The localized force pairs separated the response into distinct component patterns (Fig. 3). Both transverse pairs reduced the absolute CC component relative to bilateral support, but the ML component increased only under LAB2: its paired median contrast was $+0.052$ mm (95% CI $+0.047$ to $+0.061$, $q < 0.001$), compared with -0.007 mm for LAB1 (95% CI -0.018 to $+0.002$, $q = 0.254$). Both also reduced the AP component. LAB3 increased the ML and AP components while reducing the CC component. The shared transverse-redistribution prediction was therefore only partly supported. Supplementary Table S1 gives all paired contrasts and

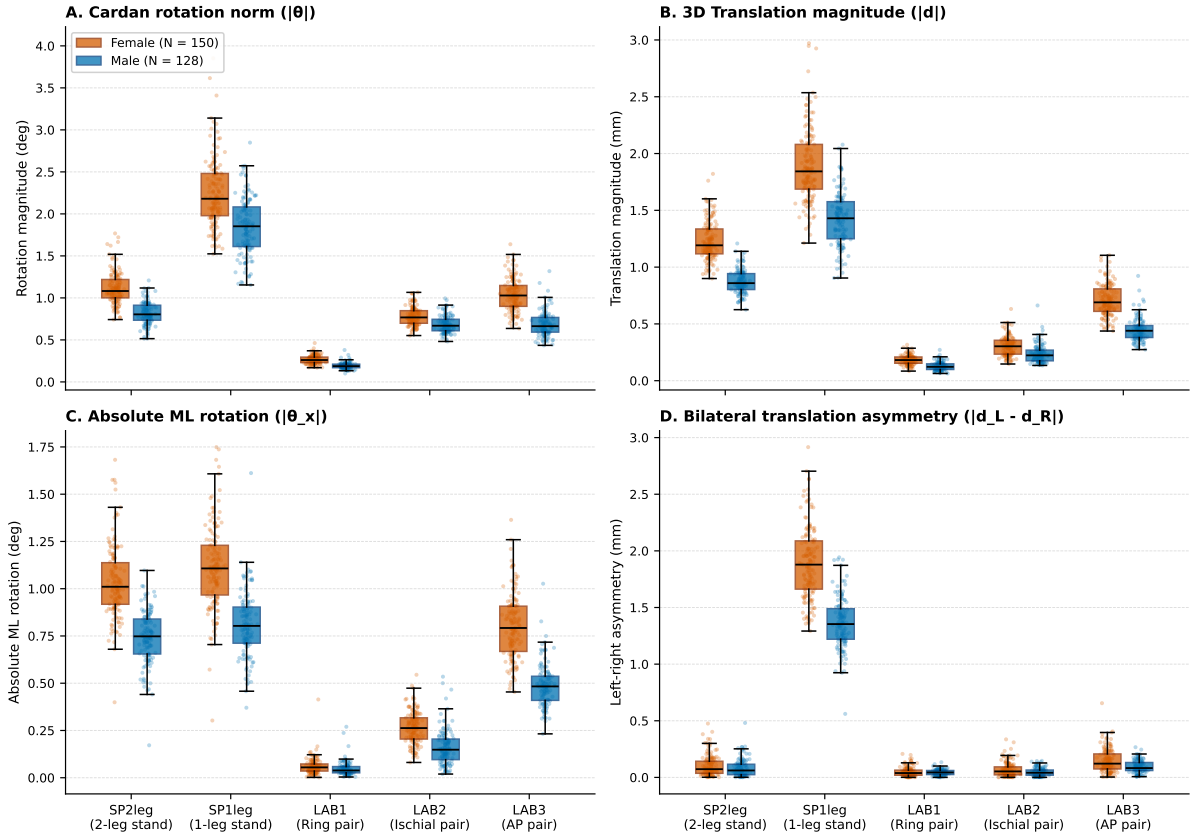


Figure 2: **Load-dependent motion and asymmetry.** Rotation-triplet norm, relative translation norm, absolute ML rotation and bilateral translation-magnitude asymmetry. Points are subjects; boxplots show medians and IQRs, with whiskers at 1.5 IQR. Female and male groups use vermilion and blue. Jitter is deterministic. No signed nutation direction is inferred.

adjusted probabilities. Component magnitudes quantify motion along the reporting axes; the arrows connect componentwise median contrasts and are not physical trajectories.

3.4 Pelvic dimensions attenuate sex-associated translation differences

LAB1 female-minus-male translation coefficients were +0.055, +0.040 and +0.013 mm in M0, M1 and M2. The M2 interval was -0.005 to +0.031 mm ($p = 0.166$, $q = 0.566$). LAB2 female-minus-male translation coefficients were +0.071, +0.041 and -0.031 mm in M0, M1 and M2. The M2 interval was -0.078 to +0.016 mm ($p = 0.192$, $q = 0.566$). LAB3 female-minus-male translation coefficients were +0.258, +0.212 and +0.002 mm in M0, M1 and M2. The M2 interval was -0.060 to +0.064 mm ($p = 0.950$, $q = 0.950$). The maximum non-intercept VIF in M2 was 5.55. For LAB1, the pooled Spearman correlation between subpubic angle and translation magnitude was $r_s = 0.444$ ($p < 0.001$). The male-only correlation was $r_s = -0.063$ ($p = 0.480$). The female-only correlation was $r_s = -0.145$ ($p = 0.077$). These exploratory correlations are uncorrected for multiplicity and do not estimate a geometric mechanism. Figure 4 and Supplementary Table S3 report M2 estimates for both outcomes.

3.5 Individual geometry preserves responses under standardized bone density

Across five loads and five endpoints, shape-only/full variance ratios ranged from 96.43% to 120.61% (median 100.68%). Material-only/full ratios ranged from 0.014% to 5.789% (median 0.244%). For rotation, minimum shape/full identity-line R^2 was 0.8777, and maximum RMSE was 0.0318 degrees. For translation, minimum shape/full identity-line R^2 was 0.9457, and maximum RMSE was 0.0280 mm. These compare model variants and do not decompose an additive population variance budget (Fig. 5; Supplementary Table S2).

3.6 Larger pelves show smaller motions across loading configurations

20 of 20 sex-specific slopes survived FDR correction. The exponents ranged from -1.062 to -0.366; the largest adjusted slope q was 0.002. The inverse association with pelvic volume was therefore consistent across all five loading configurations. Full estimates and intervals are in Table 4. Sex-by-volume interaction q values ranged from 0.343 to 0.699; nonsignificance does not establish equivalence. Figure 6 and Table 4 use the same adjusted models. Pooled cluster-robust coefficients are reported in Supplementary Table S5.

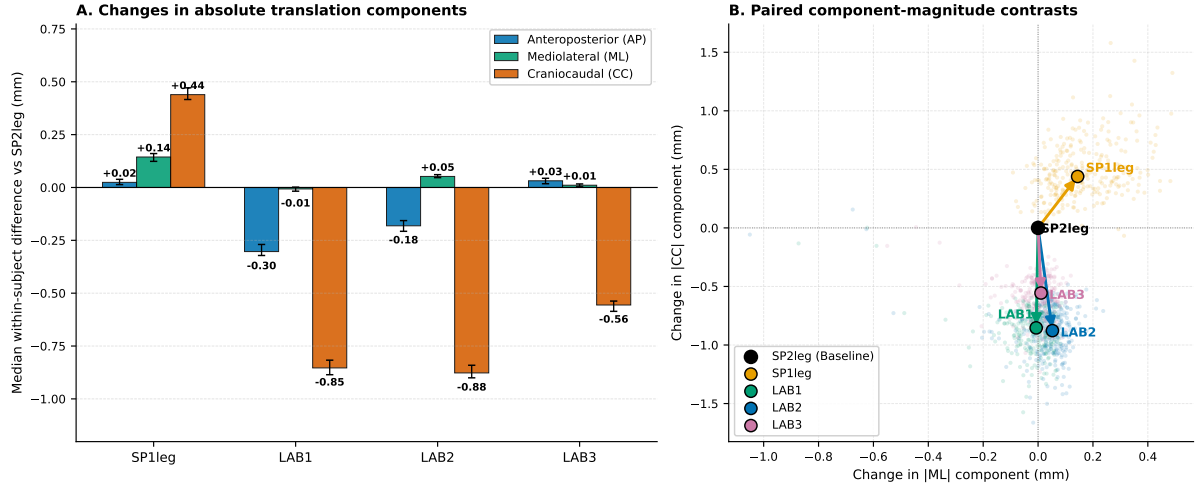


Figure 3: **Paired absolute-component contrasts.** (A) Median within-subject differences from SP2leg, with bootstrap 95% CIs. (B) Individual ML/CC contrast pairs and arrows to their componentwise medians. Positive values mean larger component magnitude, not outward joint motion. Corrected q values are reported in Supplementary Table S1; intervals are pointwise, not simultaneous.

Table 4: Adjusted log-log exponents by load and sex. Slope q values use the 20-test family; interaction q values use a separate ten-test family.

Load	Outcome	Male exponent [95% CI]	Male q	Female exponent [95% CI]	Female q	Interaction q
SP2leg	Rotation (deg)	-0.679 [-0.890, -0.469]	< 0.001	-0.605 [-0.764, -0.446]	< 0.001	0.699
SP2leg	Translation (mm)	-0.366 [-0.531, -0.200]	< 0.001	-0.402 [-0.522, -0.282]	< 0.001	0.699
SP1leg	Rotation (deg)	-0.622 [-0.895, -0.349]	< 0.001	-0.550 [-0.760, -0.339]	< 0.001	0.699
SP1leg	Translation (mm)	-0.460 [-0.711, -0.208]	< 0.001	-0.397 [-0.591, -0.203]	< 0.001	0.699
LAB1	Rotation (deg)	-0.916 [-1.144, -0.687]	< 0.001	-1.062 [-1.233, -0.891]	< 0.001	0.588
LAB1	Translation (mm)	-0.568 [-0.925, -0.210]	0.002	-1.003 [-1.270, -0.737]	< 0.001	0.343
LAB2	Rotation (deg)	-0.541 [-0.714, -0.367]	< 0.001	-0.674 [-0.804, -0.544]	< 0.001	0.494
LAB2	Translation (mm)	-0.818 [-1.179, -0.456]	< 0.001	-0.947 [-1.287, -0.606]	< 0.001	0.699
LAB3	Rotation (deg)	-0.850 [-1.048, -0.651]	< 0.001	-1.045 [-1.240, -0.851]	< 0.001	0.494
LAB3	Translation (mm)	-0.700 [-0.922, -0.478]	< 0.001	-0.904 [-1.159, -0.649]	< 0.001	0.494

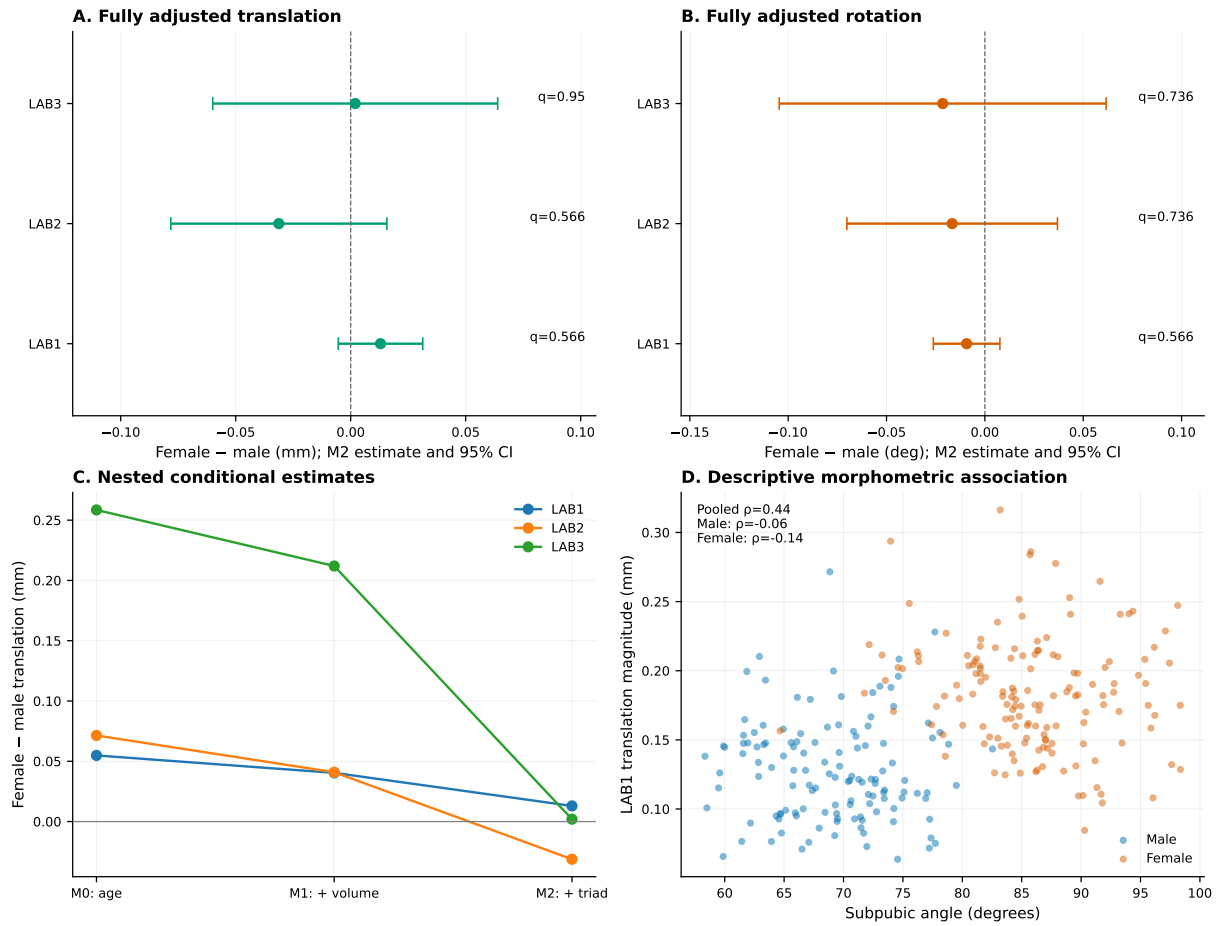


Figure 4: **Pelvic dimensions reshape the sex comparison.** (A,B) M2 female-minus-male coefficients with HC3 95% CIs, shown on separate translation and rotation axes. Labels give BH-adjusted q values across six tests. (C) Nested translation coefficients; connecting lines show changes between model specifications, not causal pathways. (D) LAB1 translation versus subpubic angle, with pooled and sex-stratified Spearman correlations. M0 adjusts for age, M1 adds volume, and M2 adds the morphometric triad.

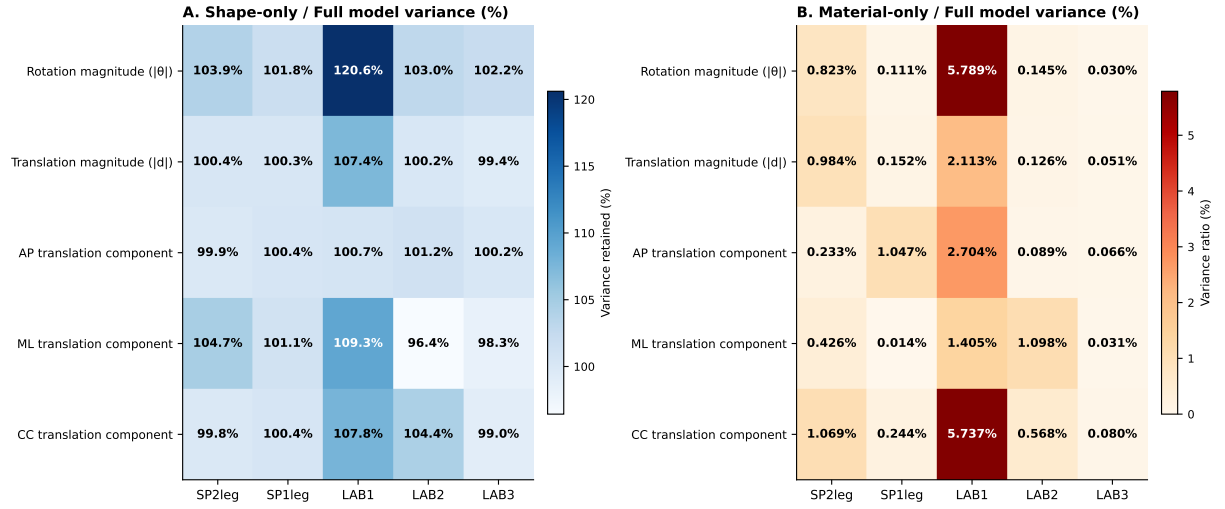


Figure 5: **Variance ratios across matched variants.** (A) Shape-only/full and (B) material-only/full sample-variance ratios, in percent. Five endpoints are shown. Color scales differ between panels and follow their actual data ranges. Values above 100% are allowed. Bootstrap intervals and paired agreement diagnostics are provided in Supplementary Table S2; neither panel represents variance explained by an independent causal factor.

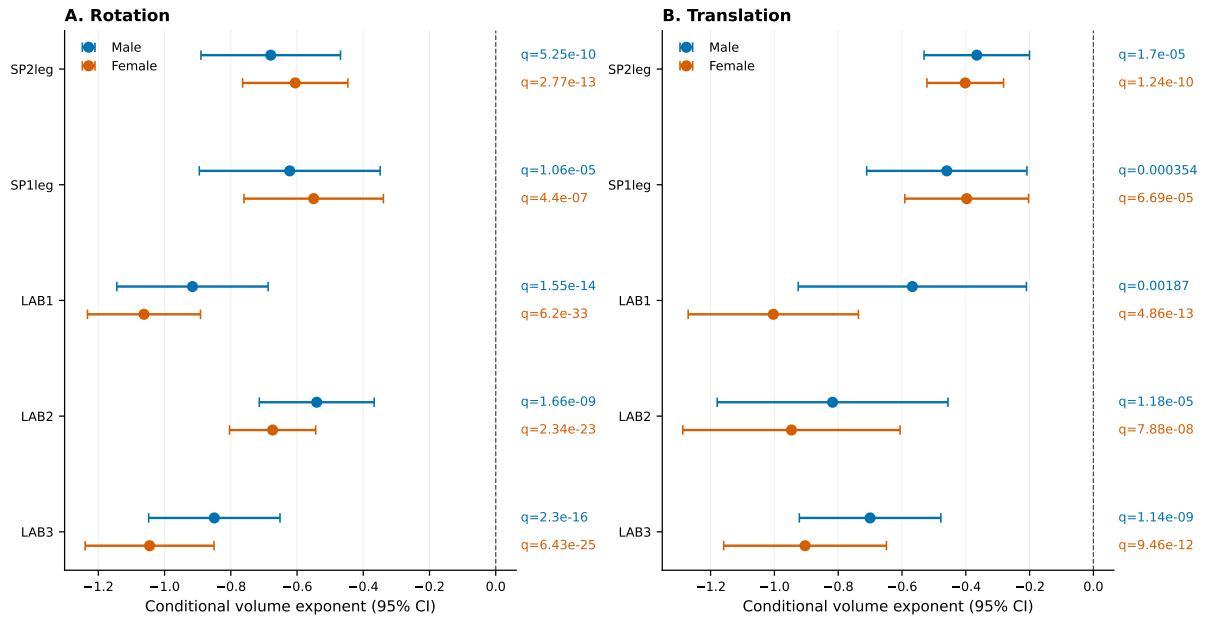


Figure 6: **Conditional log-volume slopes.** Male and female exponents with HC3 95% CIs from the same adjusted log-log models as Table 4. The vertical zero line indicates no conditional association. Confidence intervals are pointwise; labels give slope q values from the 20-test family. Raw scatterplots and adjusted conditional geometric-mean curves are supplied as Supplementary Figure S1.

3.7 Sensitivity and synthesis of the hypothesis tests

In six subjects selected at approximately evenly spaced archive indices (30 subject–load comparisons), replacing affine sacral correction with rigid-only correction changed bilateral rotation norms by at most 0.0229° and translation magnitudes by at most 0.0055 mm. This subset diagnostic does not bound errors in the remaining cohort (Supplementary Table S6).

- **H1:** Supported for the specified static endpoints. This does not verify a nutation or force-closure mechanism.
- **H2:** The full proposed LAB1/LAB2 component pattern is not supported after multiplicity correction. LAB3 is a separate contrast; signed opening remains untested.
- **H3:** The nested models support the proposed attenuation of age-adjusted female translation differences after inclusion of size and pelvic dimensions. All three fully adjusted translation intervals include zero; the coefficient changes do not establish causal mediation.
- **H4:** Shape/full variance retention and paired errors support only the stated comparison with a reference bone-density field. They do not rank geometry against unmodeled soft-tissue variability.

4 Discussion

Pelvic mobility emerges here as a response to both architecture and loading. Redistributing the same total acetabular force from two sides to one increased the median bilateral translation asymmetry from 0.065 to 1.602 mm. Localized force pairs elicited distinct component patterns, showing that their effects cannot be summarized as a uniform increase in mobility. Across these configurations, larger pelves showed smaller motions, sex-associated translation differences attenuated after adjustment for pelvic dimensions, and individual geometry preserved much of the response when bone density was standardized. These findings connect population variation in anatomy to the way the pelvic ring responds to different mechanical demands.

Load distribution selects the joint response

The bilateral–unilateral comparison isolates a simple but consequential change: an 800 N total force is redistributed without increasing its magnitude. Median translation increased from 1.058

to 1.627 mm and median rotation from 0.956 to 2.040°, while asymmetry increased much more strongly. The ratio of cohort median asymmetries was approximately 25; this is a comparison of group summaries, not a typical individual fold change. Thus total force alone is a poor description of the mechanical challenge to the SIJs. Its distribution determines moments and load paths through the ring.

The localized probes further show why the direction and location of loading must be considered together. Both transverse force pairs reduced the craniocaudal component relative to bilateral standing, but only the ischial pair increased the mediolateral component after multiplicity correction. The anteroposterior pair produced another combination of component changes. The complete transverse-redistribution hypothesis was therefore not supported: two force pairs acting along the same global axis did not produce interchangeable responses. This result makes application site a substantive anatomical variable, rather than a detail of the loading protocol.

These comparisons complement established accounts of SIJ load transfer (Snijders et al., 1993; Vleeming et al., 2012). They identify which loads accentuate asymmetry and which separate motion components, providing specific configurations for subsequent experimental testing. Because the reported components are absolute values in a pelvis-aligned frame, they describe the distribution of motion rather than signed nutation or local joint opening. Obstetric relevance lies in the choice of anatomical probes; birth-canal expansion itself was not measured.

Anatomical dimensions sharpen the interpretation of sex differences

The sex comparison becomes more informative when considered together with pelvic dimensions. Age-adjusted female translation coefficients were positive under all three localized loads, ranging from 0.055 to 0.258 mm. After including volume, anteroposterior diameter, biischiadic width and subpubic angle, the coefficients ranged from -0.031 to $+0.013$ mm, with all intervals spanning zero. The data therefore support a substantial attenuation of the initial group differences when measured anatomy is described explicitly. They do not establish mechanical equivalence between sexes or causal mediation by these dimensions.

The subpubic-angle comparison illustrates why this distinction matters. Its positive pooled association with LAB1 translation ($r_s = 0.444$) was absent within either sex. A population association can consequently suggest a simple shape-motion rule that does not hold within the constituent groups. Rather than treating sex as a sufficient mechanical descriptor, the findings motivate analyses of specific dimensions and their combinations. Prediction in new individuals

will require external testing, including assessment of the range over which male and female anatomies overlap.

Individual geometry preserves responses under standardized bone density

The matched variants offer a second route to understanding anatomical variation. With each individual's geometry retained and bone density replaced by a spatially varying population-median field, translation RMSE remained at or below 0.028 mm across loads; rotation RMSE remained at or below 0.032°. Minimum identity-line R^2 values were 0.946 for translation and 0.878 for rotation. These paired comparisons show that the reference density field preserved individual responses reasonably closely for the reported endpoints, with better agreement for translation than rotation.

The population comparison is consistent with this result: shape-only/full variance ratios had a median of 100.68%, whereas the corresponding material-only/full median was 0.244%. Ratios above 100% are possible because the variants are separate mechanical systems, not terms in an additive variance decomposition. The practical implication is that retaining individual geometry may be a useful priority when constructing models for these joint-motion outputs. Whether density standardization is acceptable in a particular application depends on the error tolerance and endpoint.

This conclusion concerns bone-density variation under the implemented constitutive law. The low-density modulus floor reduces sensitivity to part of the density range, and cartilage and ligament properties were held fixed. Ligament pretension can influence SIJ predictions (Heyland et al., 2025); its population variability could alter the relative importance of anatomy and tissue properties. The present comparison therefore motivates targeted simplification and sensitivity studies, rather than a general claim that material properties are unimportant.

Size contributes consistently across distinct load paths

All 20 sex-specific volume slopes were negative after multiplicity correction. Their consistency across five loads contrasts with the load-specific component patterns: pelvic size was associated with the magnitude of motion even as the distribution of motion changed. This provides a common structural feature across otherwise different mechanical responses.

A dimensional benchmark helps interpret that consistency. For geometrically similar linear-elastic solids under fixed force and modulus, $u \propto L^{-1}$ and $\theta \sim u/L \propto L^{-2}$, giving volume

exponents of $-1/3$ and $-2/3$. The present jointed, heterogeneous pelves also have fixed spring parameters and varying proportions, so these benchmarks are not exact predictions. Moreover, the fitted volume coefficients condition on other dimensions and describe residual size associations. Their range, -1.062 to -0.366 , should consequently be read as load- and endpoint-specific behavior under prescribed force, rather than a universal biological scaling law. Sex-by-volume interactions were not detected; equality of slopes was not tested.

Contribution and next experimental steps

The contribution beyond previous population work on SIJ morphology (Henyš and Hammer, 2025) is the connection between paired load responses, whole-pelvis dimensions and matched geometry–density comparisons. The same anatomies are tested under different loads, and the same loads are tested after changing the source of individual variation. Together, these comparisons reveal why a single mobility value or an unadjusted sex contrast misses important features of pelvic mechanics.

The findings suggest focused next experiments: compare bilateral and unilateral support at matched total force, test transverse loads at distinct application sites, and assess whether density standardization preserves measured joint motion. They also motivate direct measurements of signed surface separation and canal dimensions when studying obstetric accommodation. Such measurements would connect the component responses reported here to the functional quantities needed for childbirth models (Chen and Grimm, 2021).

Scope and limitations

These are predictions of passive, static mechanics. The fixed sacrum, nominal cartilage properties, linearized ligament representation and absence of muscle activation and frictional contact constrain physiological interpretation. The cohort spans ages 16–91 and does not represent laboring women specifically; pregnancy-related adaptation, parity, pain and delivery outcomes were not analyzed. The results establish neither clinical thresholds nor an evolutionary trade-off.

Motion was measured relative to each subject’s unloaded anatomy, keeping shape variation separate from elastic displacement. Affine sacral correction also removes sacral strain; replacing it with rigid-only correction changed translation by at most 0.0055 mm and rotation by 0.0229° in the six-subject sensitivity subset. This is an internal check, not a cohort-wide error bound. Sampling intervals likewise do not include uncertainty in constitutive parameters, mapping or

boundary conditions. Quantitative physiological use requires mesh and mapping-quality assessment, constitutive sensitivity and independent experimental validation with matched loads and endpoint definitions, especially given the difficulty of measuring small SIJ motions (Goode et al., 2008).

5 Conclusion

Pelvic architecture and load distribution jointly shape sacroiliac motion. At unchanged total force, unilateral support markedly amplified bilateral asymmetry; localized loads selected distinct motion-component patterns. Sex-associated translation differences attenuated when pelvic dimensions were included, while larger pelves showed smaller responses across all five loads. Preserving individual geometry with a reference bone-density field retained much of the individual response and population variation. These controlled comparisons provide an anatomical basis for interpreting pelvic mobility and concrete targets for experiments that connect joint motion to pelvic function.

A Mathematical formulation and verification

A.1 Reference mapping and equilibrium

Let Ω_0 be the FE template and $\chi_i(\mathbf{X}) = \mathbf{X} + \phi_i(\mathbf{X})$ the unloaded subject geometry. The anatomical mapping and the elastic displacement are different fields. With $\mathbf{F}_i = \mathbf{I} + \nabla_X \phi_i$ and $J_i = \det \mathbf{F}_i$, the physical small-strain tensor is

$$\boldsymbol{\varepsilon}_i(\mathbf{u}) = \text{sym}(\nabla_X \mathbf{u} \mathbf{F}_i^{-1}). \quad (5)$$

The pullback requires an orientation-preserving, nonsingular map. RBF smoothness alone does not prove this condition. For physical coordinates in mm, the discrete weak problem is

$$\int_{\Omega_i} \mathbf{C}_i \boldsymbol{\varepsilon}(\mathbf{u}) : \boldsymbol{\varepsilon}(\mathbf{v}) dV + \int_{\Gamma_{S1,i}} \gamma \mathbf{u} \cdot \mathbf{v} dS + \mathbf{v}_h^T \mathbf{K}_{lig,i} \mathbf{u}_h = \int_{\Gamma_{load,i}} \mathbf{t} \cdot \mathbf{v} dS + \mathbf{v}_h^T \mathbf{f}_{pre,i}. \quad (6)$$

The surface penalty has units N/mm³, because its associated traction is $\gamma \mathbf{u}$ in N/mm². It imposes an approximate fixed boundary, not exact $\mathbf{u} = 0$. The ligament stiffness and pretension terms are required for consistency with the assembled solver. For one spring segment of length

376 L , direction $\mathbf{e}$, assigned stiffness k and pretension T , its directional stiffness is

$$377 \quad \mathbf{W} = k\mathbf{e}\mathbf{e}^T + (T/L)(\mathbf{I} - \mathbf{e}\mathbf{e}^T). \quad (7)$$

378 This matrix is assembled with opposite off-diagonal blocks at its two endpoints. It is a linearized
379 spring representation without tension-only switching.

380 The surface measure obeys $dS_i = J_{surf,i}dS_0$, where

$$381 \quad J_{surf,i} = J_i \|\mathbf{F}_i^{-T} \mathbf{n}_0\|, \quad \mathbf{t} = \frac{\mathbf{F}^{tot}}{\int_{\Gamma_{load,0}} J_{surf,i} dS_0}. \quad (8)$$

382 Thus the applied physical force integral equals $\mathbf{F}^{tot}$. The penalty and load integrals use the
383 corresponding mapped surface measures. The force vectors are listed in Table 2; their global
384 axes are not assumed identical to the PCA reporting axes below.

385 A.2 Unloaded and loaded surfaces

386 Let $\mathbf{X}_j$ be surface template nodes. The two point clouds used for kinematics are

$$387 \quad \mathbf{p}_{ij}^0 = \mathbf{X}_j + \phi_i(\mathbf{X}_j), \quad \mathbf{p}_{ilj}^1 = \mathbf{p}_{ij}^0 + \mathbf{u}_{il}(\mathbf{X}_j). \quad (9)$$

388 Both use identical topology and point correspondence. Comparing $\mathbf{X}_j$ with $\mathbf{p}_{ilj}^1$ would incor-
389 rectly include subject morphology in the measured motion. Geometry was reconstructed with
390 the original two-stage interpolation: template-to-FE mapping (10 neighbors, smoothing 50) fol-
391 lowed by FE-to-surface mapping (10 neighbors, zero smoothing). Saved elastic fields use the
392 latter interpolation. Cached sparse interpolation weights implement the same linear RBF eval-
393 uation; numerical comparisons against direct RBF evaluation check this equivalence.

394 A.3 Pelvis-aligned reporting frame

395 The origin $\mathbf{o}$ is the centroid of the unloaded surface. The initial ML axis is the dominant
396 eigenvector of the point-cloud covariance, with deterministic sign. The symmetry-plane routine
397 uses this PCA approximation; it does not optimize reflection symmetry. Project the centered
398 points onto its orthogonal plane and take the leading in-plane singular vector. Orthogonalization
399 gives $\hat{\mathbf{z}}$, followed by

$$400 \quad \hat{\mathbf{y}} = \frac{\hat{\mathbf{z}} \times \hat{\mathbf{x}}}{\|\hat{\mathbf{z}} \times \hat{\mathbf{x}}\|}, \quad \hat{\mathbf{z}} = \hat{\mathbf{x}} \times \hat{\mathbf{y}}, \quad \mathbf{P} = [\hat{\mathbf{x}}, \hat{\mathbf{y}}, \hat{\mathbf{z}}]. \quad (10)$$

The ML and AP axes are simultaneously reversed if needed so that ML points from the left to right ilium centroids. This construction is orthonormal and right-handed but does not independently anchor the AP/CC signs to anatomical landmarks. Absolute components are therefore used for comparisons. No clinical nutation sign is inferred from the stored α_x sign.

A.4 Affine correction, regions and rigid fits

Fit the sacral point correspondence by ordinary least squares,

$$(\mathbf{A}_S, \mathbf{b}_S) = \arg \min_{\mathbf{A}, \mathbf{b}} \sum_{j \in S} \|\mathbf{p}_j^1 - \mathbf{A} \mathbf{p}_j^0 - \mathbf{b}\|^2, \quad \mathbf{q}_j = \mathbf{A}_S^+(\mathbf{p}_j^1 - \mathbf{b}_S), \quad (11)$$

where $+$ denotes the Moore–Penrose pseudoinverse. This removes fitted affine sacral deformation, including stretch and shear; it is not only a change of rigid observer frame.

For each side, select sacral and iliac points within 5 mm of the opposing point cloud. If fewer than 80 points qualify, use up to the nearest 80 available points. Distances are between discrete points rather than continuous articular surfaces. For each bone ROI, the rigid fit maps $\mathbf{p}^0$ to $\mathbf{q}$. With centered row matrices $\tilde{\mathbf{P}}_B, \tilde{\mathbf{Q}}_B$,

$$\tilde{\mathbf{P}}_B^T \tilde{\mathbf{Q}}_B = \mathbf{U} \Sigma \mathbf{V}^T, \quad \mathbf{R}_B = \mathbf{V} \text{diag}(1, 1, \det(\mathbf{V} \mathbf{U}^T)) \mathbf{U}^T, \quad \mathbf{t}_B = \bar{\mathbf{q}}_B - \mathbf{R}_B \bar{\mathbf{p}}_B^0. \quad (12)$$

The fit is repeated after excluding the largest 10% residuals, using two trimming iterations. Trimming reduces outlier influence but does not validate the ROI as a physical contact patch.

A.5 Relative rotation and translation

For sacral and iliac fits S, I ,

$$\mathbf{R}_{rel} = \mathbf{R}_S^T \mathbf{R}_I, \quad \mathbf{R}_{loc} = \mathbf{P}^T \mathbf{R}_{rel} \mathbf{P} = \mathbf{R}_z(\gamma_z) \mathbf{R}_y(\beta_y) \mathbf{R}_x(\alpha_x). \quad (13)$$

The fixed-axis xyz extraction, for zero-based matrix indices away from gimbal lock, is

$$\alpha_x = \text{atan2}(R_{21}, R_{22}), \quad \beta_y = \arcsin[-\text{clip}(R_{20}, -1, 1)], \quad \gamma_z = \text{atan2}(R_{10}, R_{00}). \quad (14)$$

At $\beta_y = \pm\pi/2$, choose $\gamma_z = 0$ and $\alpha_x = \text{atan2}[-\text{sign}(R_{20})R_{01}, R_{11}]$. In implementation, the equivalent stable expression $\beta_y = \text{atan2}(-R_{20}, \sqrt{R_{00}^2 + R_{10}^2})$ is used, with the singular branch

restricted to vanishing cosine. The singular branch is included for completeness; physiological small-angle outputs should remain far from it. Matrix reconstruction tests cover both branches.

For the unloaded sacral ROI centroid $\mathbf{p}_c$, the reported displacement is

$$\mathbf{d} = \mathbf{P}^T[(\mathbf{R}_I - \mathbf{R}_S)\mathbf{p}_c + \mathbf{t}_I - \mathbf{t}_S]. \quad (15)$$

It evaluates the difference of two fitted maps at a common reference point. It is neither the distance between opposing articular surfaces nor a displacement along a surface normal. The bilateral summaries are

$$|\boldsymbol{\theta}| = \frac{1}{2} \sum_{s=L,R} (\alpha_{x,s}^2 + \beta_{y,s}^2 + \gamma_{z,s}^2)^{1/2}, \quad |\mathbf{d}| = \frac{1}{2} \sum_{s=L,R} \|\mathbf{d}_s\|, \quad (16)$$

$$|d_k| = \frac{1}{2}(|d_{k,L}| + |d_{k,R}|), \quad A_d = \left| \|\mathbf{d}_L\| - \|\mathbf{d}_R\| \right|. \quad (17)$$

Angles are converted to degrees before reporting. The exact principal rotation angle would instead be $\arccos[(\text{tr } \mathbf{R}_{rel} - 1)/2]$ with numerical clipping. The Cardan norm approximates it at small angles and depends on the chosen axes and sequence.

A.6 Size and conditional scaling

The archived allometry workflow sums volumes of the selected left ilium, right ilium and sacrum surface bodies, using mapped surface geometry and converting mm^3 to cm^3 . These are whole-bone geometric volumes, not birth-canal volume or a direct integral of cortical tissue. The size factor is $s_i = (V_i/V_{template})^{1/3}$. Therefore a log-volume slope b becomes $3b$ under log scale, with identical fitted responses and tests.

For the sex-interaction log model, $b_M = \beta_v$ and $b_F = \beta_v + \beta_{v:s}$. The female slope variance is

$$\text{Var}(b_F) = \text{Var}(\beta_v) + \text{Var}(\beta_{v:s}) + 2 \text{Cov}(\beta_v, \beta_{v:s}). \quad (18)$$

Reported intervals use this full covariance. Exponentiating a fitted log response gives a conditional geometric mean, not an unbiased arithmetic mean. Figure 6 uses this same adjusted model, with the other continuous covariates set to their cohort means.

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Author contributions

CRedit authorship contribution statement. Petr Henyš: Conceptualization; Methodology; Software; Formal analysis; Investigation; Data curation; Visualization; Funding acquisition; Writing – original draft; Writing – review & editing. Niels Hammer: Supervision; Writing – review & editing. Final author approval of this revised version is pending.

Conflict of interest statement

The authors declare no conflicts of interest.

Supporting information

Supporting Information provides statistical definitions, component contrasts, variance ratios, sex models, the scale reparameterization, and extraction sensitivity diagnostics.

Data Availability Statement

The manuscript working package contains corrected subject-level endpoints, statistical tables and figures. The original FE displacement archives are retained separately. A persistent public or reviewer-accessible repository link must be supplied before submission.

Code Availability Statement

The working package includes scripts for corrected endpoint extraction, statistical analysis and figure generation. A versioned code archive and environment specification must accompany the

data repository before submission.

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